Protein Functional Analysis of BRCA1 Using UniProt

Aastha Mani
B.Tech. Biotechnology, Graphic Era University
Independent Learning Project

Abstract

Bioinformatics databases provide essential information for understanding protein structure, function, and biological roles. In this study, the protein sequence and functional information of the BRCA1 protein from Homo sapiens were analyzed using the UniProt. The UniProt database provides detailed information including protein sequence, functional annotation, domain architecture, and disease associations. The analysis revealed that BRCA1 plays a major role in DNA repair and tumor suppression. The protein contains several important domains responsible for protein–protein interactions and DNA damage response. This study demonstrates the usefulness of protein databases in understanding molecular functions and biological significance of proteins.

Introduction

Bioinformatics is a scientific discipline that combines the fields of biology, computer science, and information technology. Bioinformatics plays a vital role in the analysis of biological data. With the advancement of genomic and proteomic research, large databases have been developed to store biological information.

Protein databases play a crucial role in understanding biological functions and molecular mechanisms. One of the most widely used protein databases is the UniProt, which provides comprehensive information about protein sequences and annotations.

The BRCA1 protein is an important tumor suppressor protein involved in DNA repair and maintenance of genomic stability. Mutations in the BRCA1 gene are associated with increased risk of breast and ovarian cancers. Studying the protein structure and functional domains can provide insights into its biological roles.

This study aims to explore the protein information of BRCA1 using the UniProt database and to understand its functional and structural characteristics.

Learning Objective

The objective of this study was to gain practical experience in using protein databases for bioinformatics analysis. This exercise was performed as a learning activity by a second-semester undergraduate biotechnology student to understand how protein sequence databases can be used to analyze protein functions and structural features.

Materials and Methods

Protein data used in this study was obtained from the UniProt. The protein selected for analysis was:

- Protein: Breast cancer type 1 susceptibility protein
- Organism: Homo sapiens
- UniProt ID: P38398 · BRCA1_HUMAN

Workflow of the Analysis

1. Search for BRCA1 protein in the UniProt database
2. Select the reviewed UniProt entry for Homo sapiens
3. Retrieve protein sequence in FASTA format
4. Analyze functional annotation and biological roles
5. Identify protein domains and structural features

6. Examine disease associations and database cross-references
7. Document observations and screenshots

Results

The UniProt database provides comprehensive information about protein sequences and functional annotations. The BRCA1 protein entry contains detailed information regarding its structure, biological function, and role in disease.

The protein consists of approximately **1863 amino acids** and is involved in the repair of DNA double-strand breaks. The functional annotation in UniProt indicates that BRCA1 participates in maintaining genomic stability and regulating the cell cycle.

The domain analysis revealed the presence of important structural regions such as the **RING finger domain** and **BRCT domains**, which are responsible for protein–protein interactions and DNA repair mechanisms. Disease association data in UniProt indicates that mutations in BRCA1 are linked to hereditary breast and ovarian cancers.

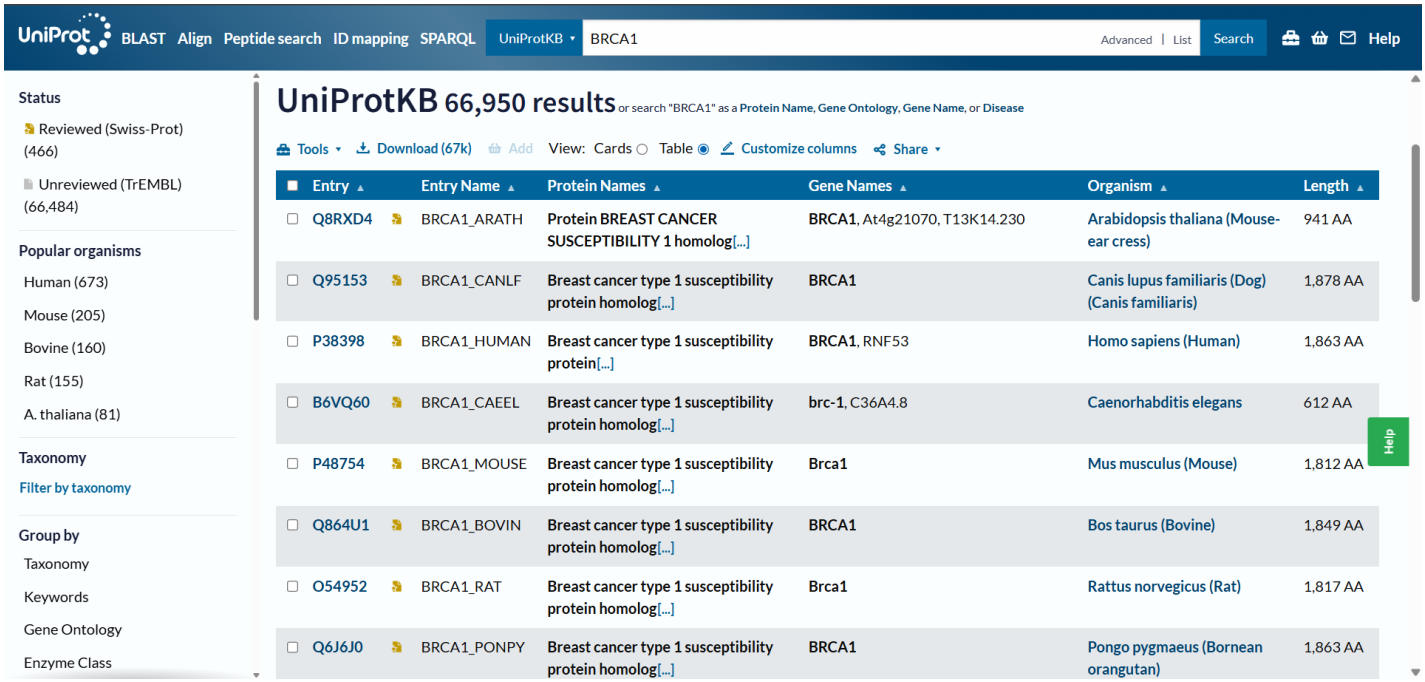

Figure 1: Search results for the BRCA1 protein in the UniProt showing multiple entries from different organisms.

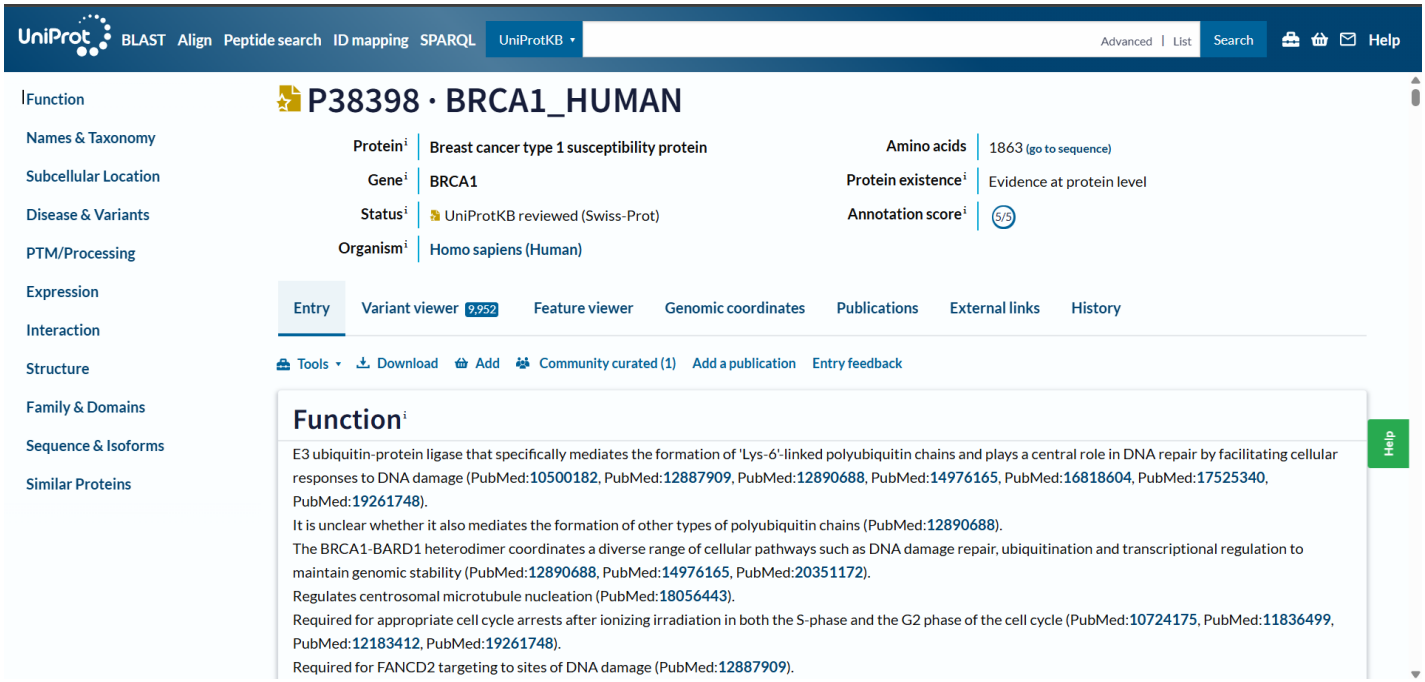

Figure 2: UniProt entry page for the BRCA1 protein (**UniProt ID: P38398**) from Homo sapiens showing protein information and functional annotation.

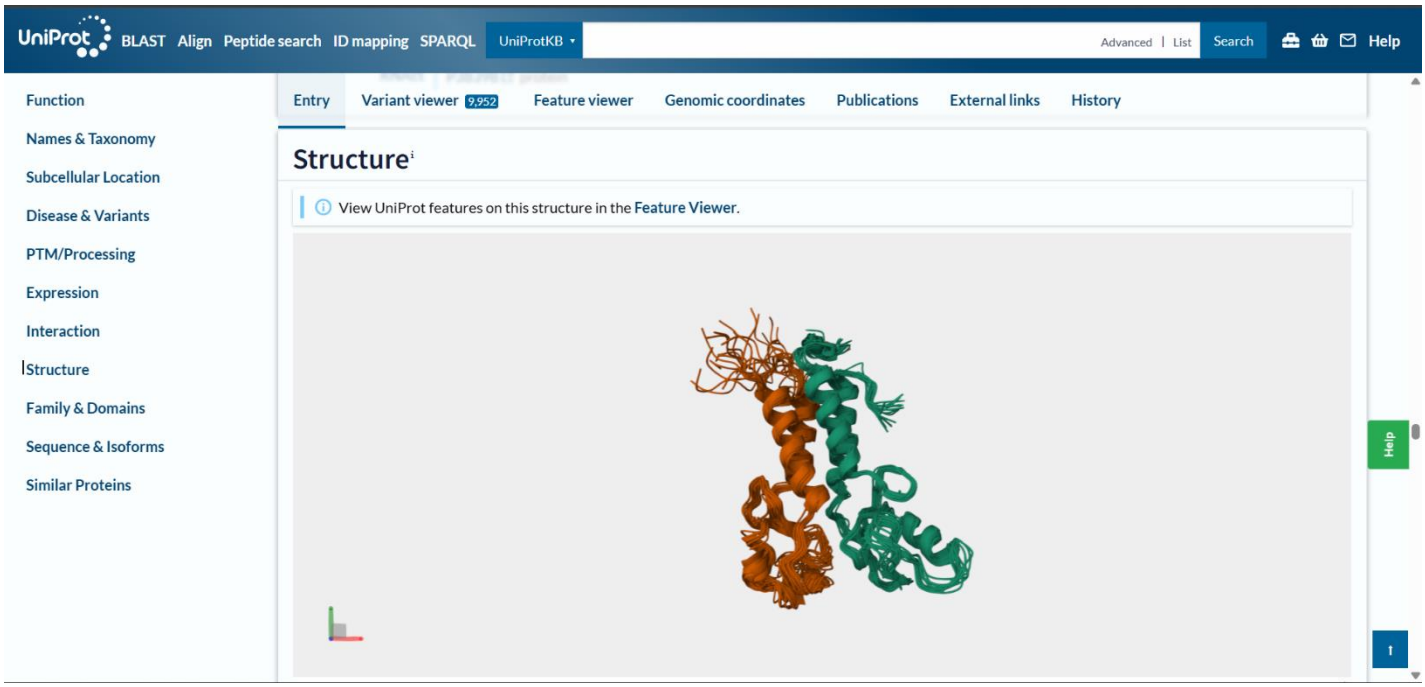

Figure 3: Three-dimensional structure of the BRCA1 protein obtained from the UniProt.

Figure 4: Feature viewer showing domains, sequence features, and annotations across the BRCA1 protein sequence.

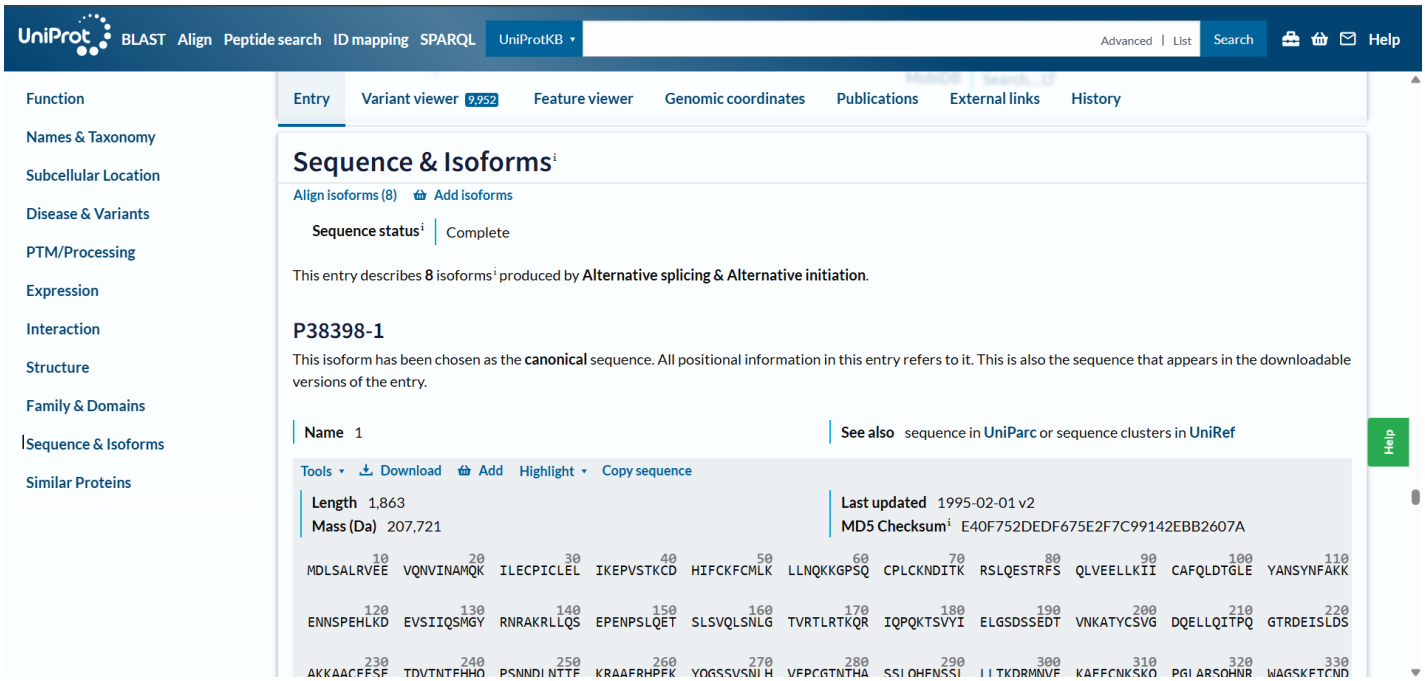

Figure 5: Amino acid sequence of the BRCA1 protein obtained from the UniProt *Sequence & Isoforms* section.

Discussion

The results of the analysis of the BRCA1 protein using the UniProt database show the significance of protein databases to the current biological research. The database provides comprehensive annotation of the protein, which is crucial to understanding the function of the protein.

The fact that the protein has conserved domains such as the RING finger domain and the BRCT domain implies the importance of the BRCA1 protein to the process of repairing damaged DNA. The fact that the protein is associated with diseases such as cancer underscores the significance of the protein to the process.

The results of the study show the importance of bioinformatics tools to the study of biological molecules and the role they play in the living cell.

Limitations of the Study

Although the analysis of the BRCA1 protein using the UniProt database provided a clear understanding of its functional properties and biological role, there are certain limitations associated with the study that need to be understood.

Firstly, it is to be noted that the analysis of the protein, as performed in this study, is entirely dependent on the information available through various biological databases. It is a known fact that biological databases, such as UniProt, are reliable sources of information. However, it is also to be noted that the analysis is entirely dependent on published data and not experimental analysis.

Secondly, it is to be noted that as performed in this study, the analysis of the protein is entirely dependent on its functional properties and not its structural properties. In other words, it is to be noted that a detailed analysis of the structural properties and behavior of the protein was not performed.

Third, the research was based on the analysis of a single protein from Homo sapiens. Comparative studies involving homologous proteins from other organisms were not included. This would have given more information on the conservation of such proteins and their similarities.

Finally, the research in this study was based on introductory studies on protein databases done for the purposes of learning. Advanced bioinformatics techniques such as protein structure prediction, molecular docking, or protein–protein interaction studies were not included in the research.

Despite these limitations, the study provides a basic understanding of how protein databases like UniProt can be used to explore protein function and biological significance in modern bioinformatics research.

Future Scope

The present study provides a basic understanding of protein analysis using biological databases such as the UniProt. However, several advanced analyses can be carried out in the future to obtain deeper insights into the biological role and structural properties of the BRCA1 protein.

Future studies could involve comparative analysis of BRCA1 protein sequences from different species to understand evolutionary conservation and functional similarities. Multiple sequence alignment tools such as Clustal Omega can be used to identify conserved regions within the protein.

Further research may also include structural analysis of the BRCA1 protein using databases such as the Protein Data Bank to study the three-dimensional structure of the protein and its functional domains. Understanding the structure of the protein can provide important insights into how mutations affect protein function.

Advanced bioinformatics approaches such as protein–protein interaction analysis, mutation impact prediction, and computational modeling could also be applied to study the functional significance of BRCA1 in DNA repair pathways and cancer development.

Overall, future research using advanced bioinformatics tools and computational techniques can provide a more comprehensive understanding of the structure, function, and biological significance of the BRCA1 protein.

Conclusion

This study provides valuable insights into the protein sequence, biological function, structural domains, and disease associations of the BRCA1 protein. In this project, the protein database information of the BRCA1 protein of Homo sapiens was studied through the UniProt database.

This project has given us valuable information about the protein sequence, biological function, domains, and diseases associated with the protein. From the database, it is clear that the protein has a vital role to play in the maintenance of genomic stability through DNA repair mechanisms and regulation of the cell cycle.

The presence of important protein domains such as the RING finger domain and the BRCT domains shows the importance of the protein in protein-protein interactions. The diseases associated with the protein also show the importance of the protein in the context of hereditary breast and ovarian cancer.

This project has given us a basic idea about protein databases. This project has shown the importance of computational tools in the field of biology. It has given us a basic idea about the importance of protein databases in the understanding of the characteristics of biological molecules without performing any experiments.

References

- UniProt Consortium
- Protein Data Bank.
- Altschul SF et al. Basic Local Alignment Search Tool (BLAST)
- National Center for Biotechnology Information Database